ATOMIC-LEVEL MATERIALS SIMULATION

Module Code: MES306TC

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Introduction

I. What is this module all about?

II. Why do we need this module?

III. How is this module delivered?

IV. Overview of the content framework of this module

I. What is this module all about?

Based on Quantum Mechanics

Study, explain and design the materials with specially required properties from the most fundamental side

Assist the design and fabrication for electronic device with the properties one interests in

I. What is this module all about?

Based on Quantum Mechanics

- Tight Binding Method
- Density Function Theory (DFT)
- Molecular Dynamics Simulation

+

- Computer Programming
- Parallelisation Technique

II. Why do we need this module?

♦ From the atomic level, explain the experimental results

♦ From the atomic level, modify the electronic properties of materials

♦ From the atomic level, predict the properties of materials

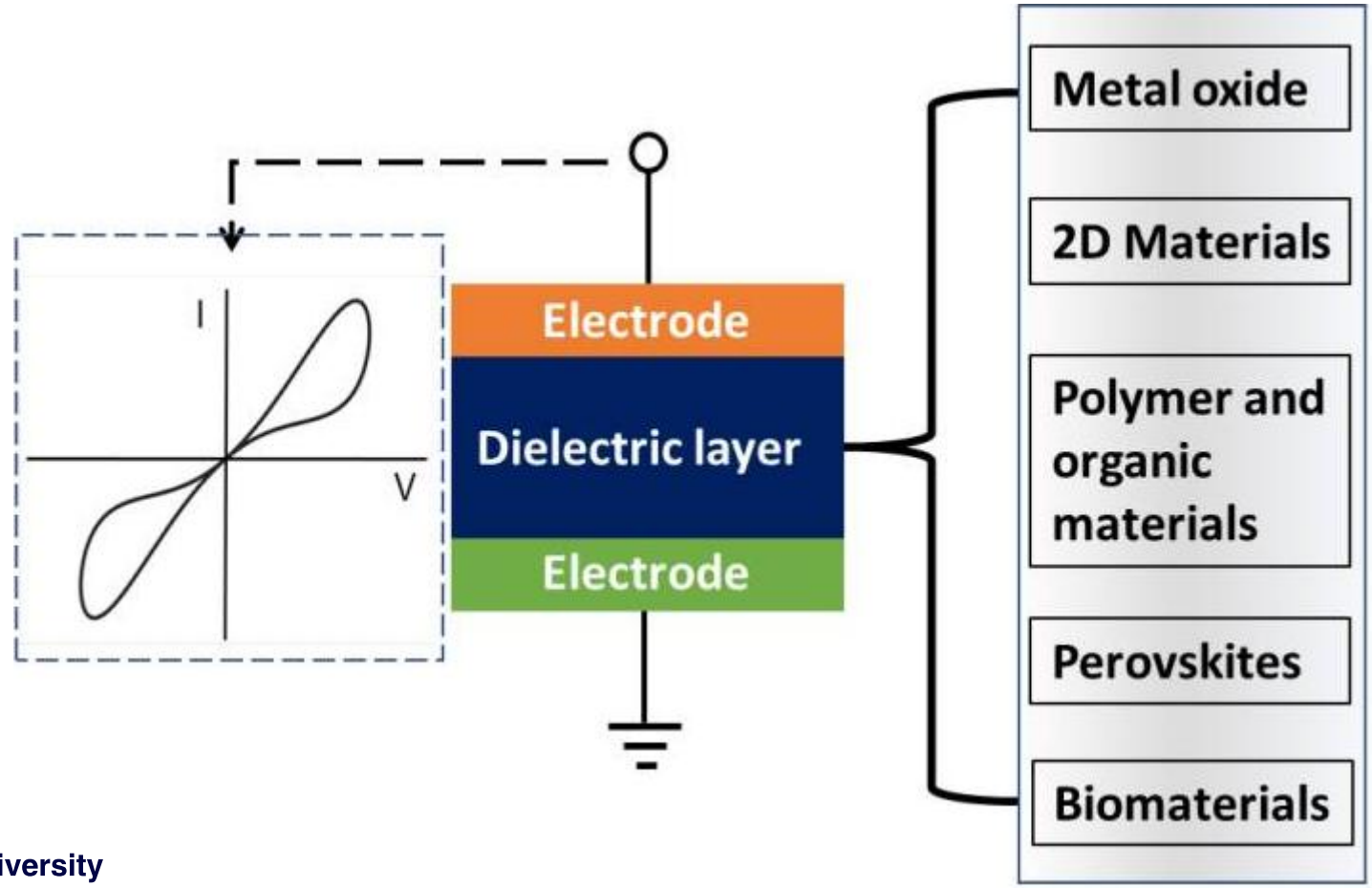
From the most fundamental side

II. Why do we need this module?

♦ From the atomic level, explain the experimental results

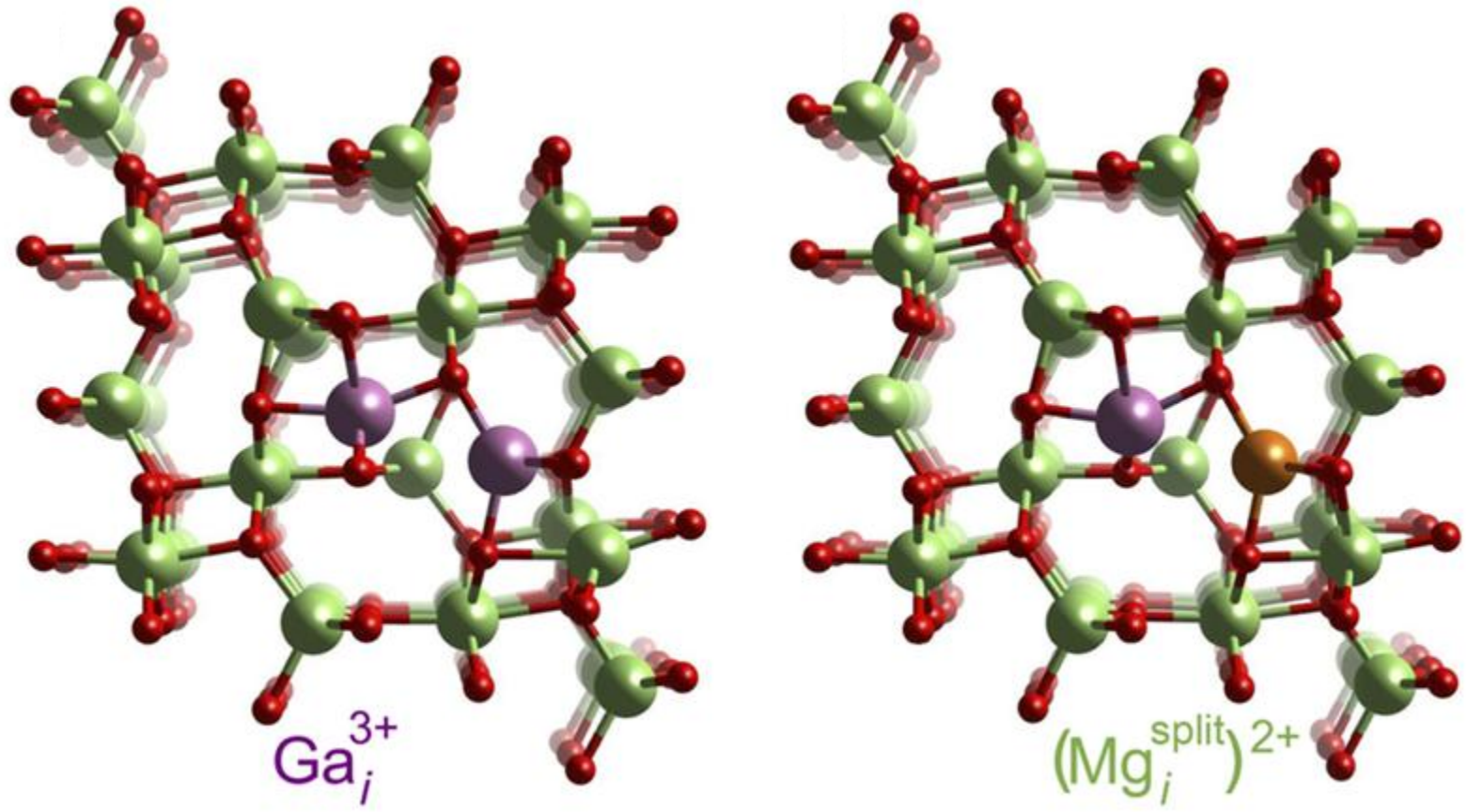
Case One: Memristor Materials Ga_2O_3

Resistance/conductance alters with direction of E field



Moore, A.; Hou, Y. and Li, L.; Gallium Oxide Memristors: A Review of Resistive Switching Devices and Emerging Applications, Nanomaterials (Basel), (2025) 15(17):1365. PMID: 40938042; PMCID: PMC12430548

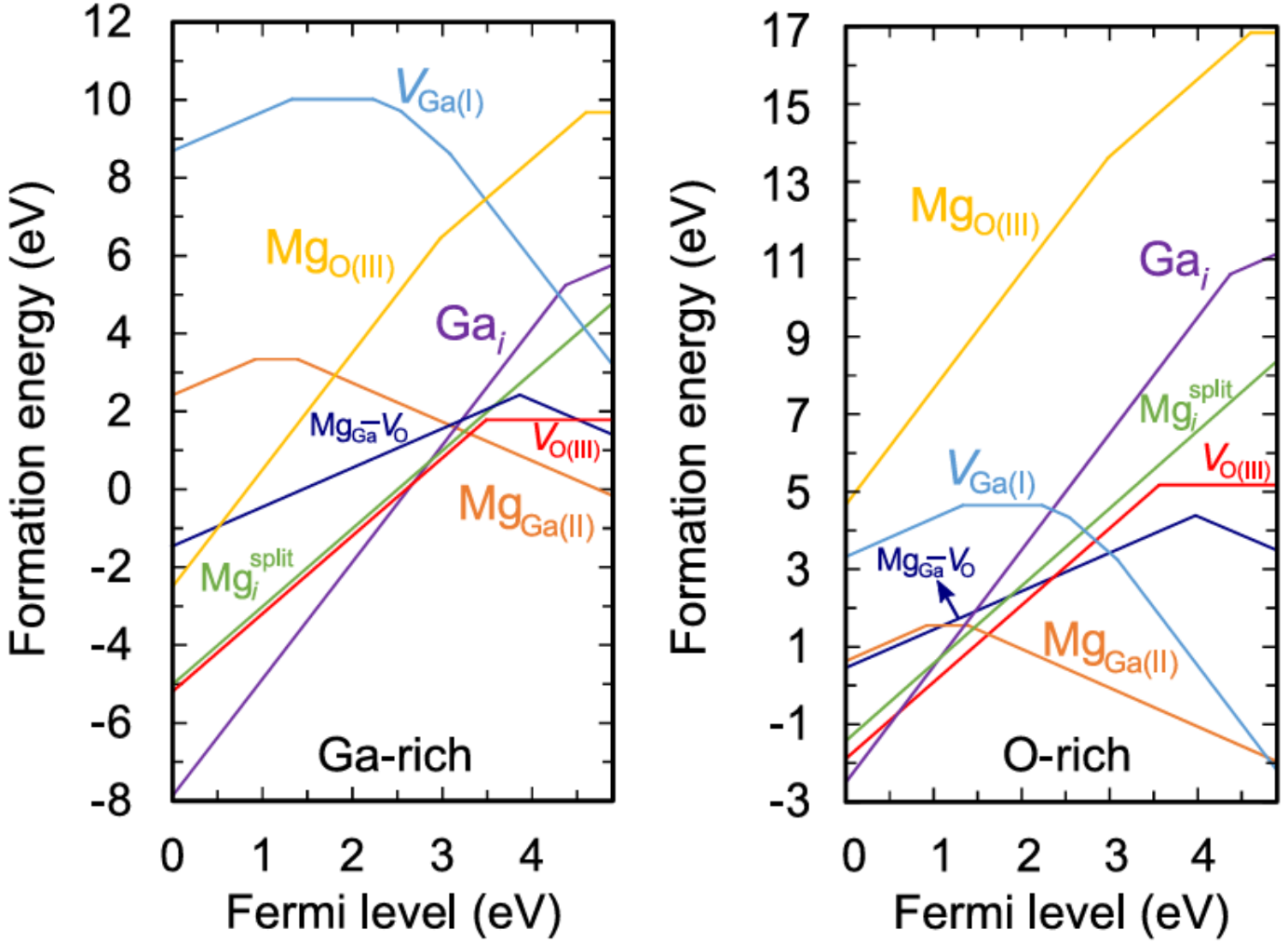
Case One: Memristor Materials Ga_2O_3



Defects, Ga^{3+}, Mg^{2+}, moves in different directions within the Ga_2O_3 lattice under E field

Hartwin Peelaers, John L. Lyons, Joel B. Varley, Chris G. Van de Walle; Deep acceptors and their diffusion in Ga_2O_3, APL Mater. (2019) 7(2):022519

Case One: Memristor Materials Ga_2O_3



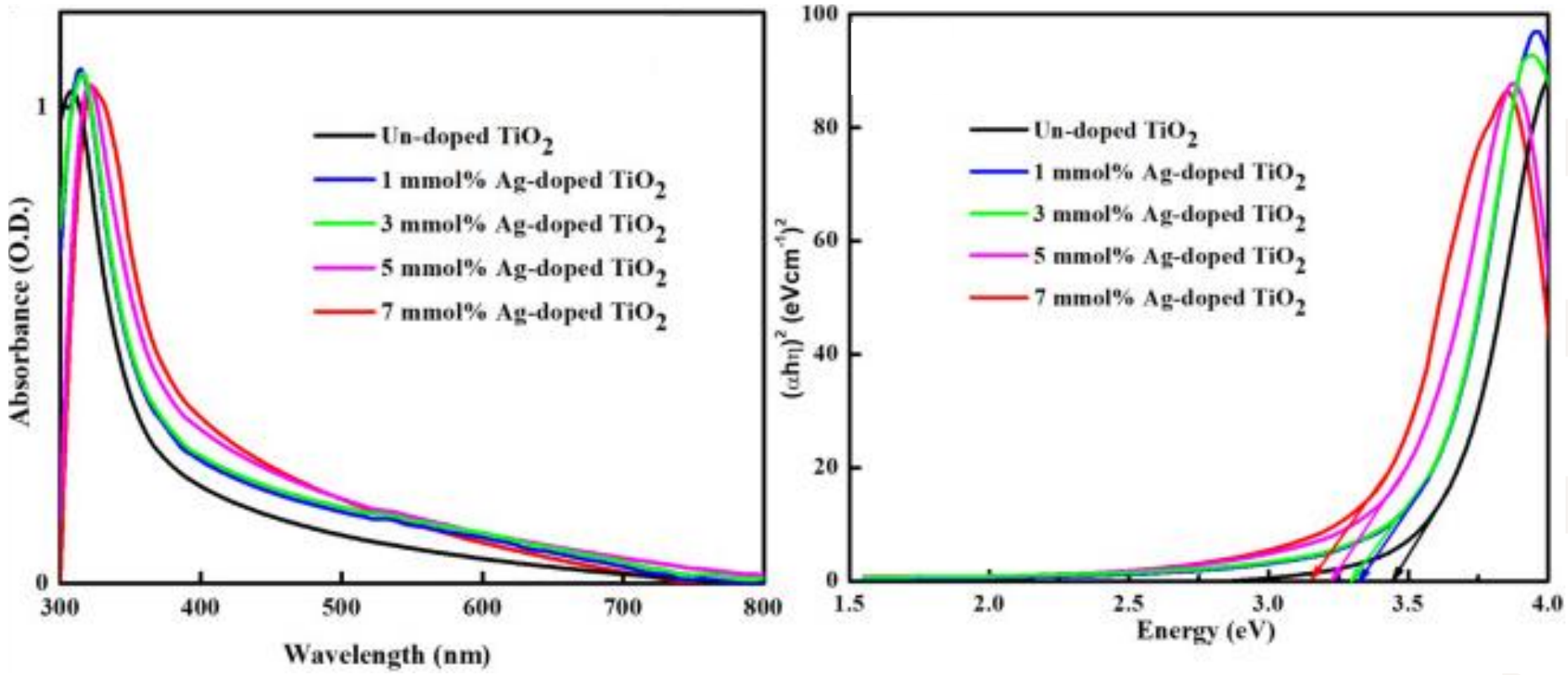
Defects moving within the lattice, Fermi level shifts which results in the charge conductance altering

Hartwin Peelaers, John L. Lyons, Joel B. Varley, Chris G. Van de Walle; Deep acceptors and their diffusion in Ga_2O_3, APL Mater. (2019) 7(2):022519

II. Why do we need this module?

♦ From the atomic level, explain the experimental results

Case Two: TiO_2 for solar energy harvesting

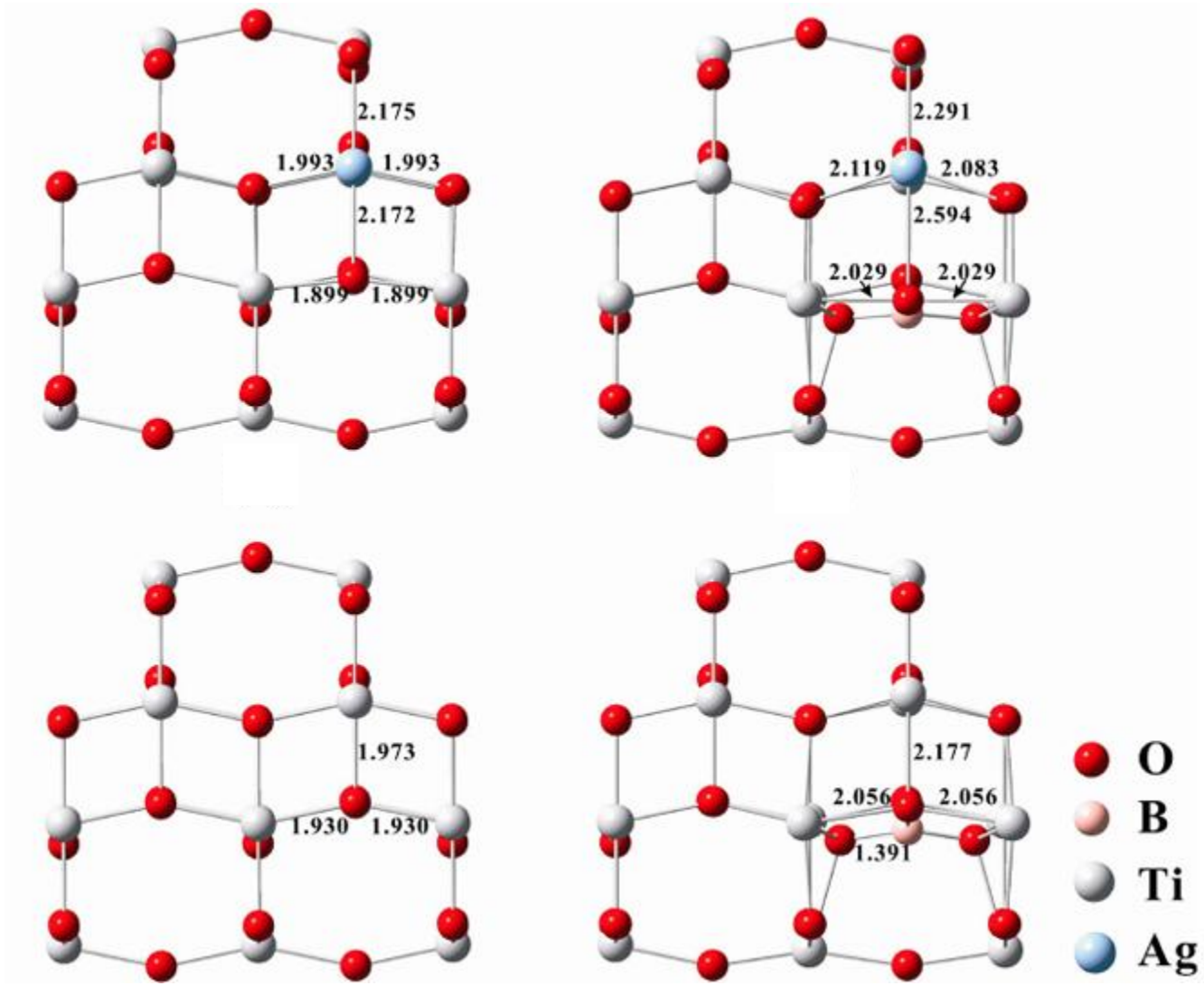


Ag doped TiO_2 has better solar energy absorbance than the pristine one

Piranave Sritharan, Meena Senthilnanthanan, Punniamoorthy Ravirajan, *et al*., Enhanced solar energy harvest in dye-sensitized solar cells using silver-doped TiO_2 Photoelectrodes via Spray Pyrolysis, Chemical Physics Impact, (2025) 11, 100910, ISSN 2667-0224

Case Two: TiO_2 for solar energy harvesting

Theoretical study of the doping effect in TiO_2



Ningdong Feng, Qiang Wang, Anmin Zheng, *et al.*, Understanding the High Photocatalytic Activity of (B, Ag)-Codoped TiO2 under Solar-Light Irradiation with XPS, Solid-State NMR, and DFT Calculations, J. Am. Chem. Soc. (2013) 135, 4, 1607–1616

Case Two: TiO_2 for solar energy harvesting

Dopant tunes the gap by altering the edge of bands

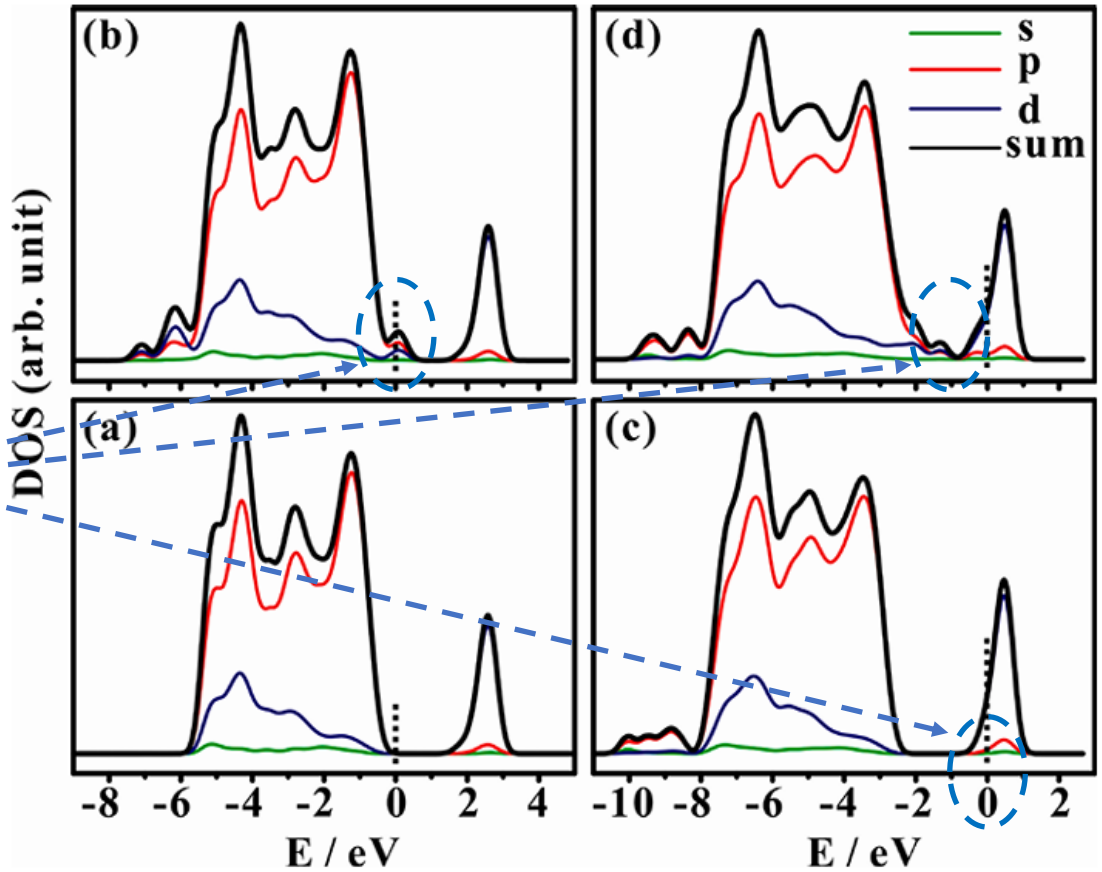


Figure 9. Density of states of (a) pure TiO_2, and doped-TiO_2 associated with (b) substitutional Ag, (c) tricoordinated interstitial boron T*, and (d) T*–O–Ag.

Xi'an Jiaotong-Liverpool University
西交利物浦大学

Ningdong Feng, Qiang Wang, Anmin Zheng, *et al.*, Understanding the High Photocatalytic Activity of (B, Ag)-Codoped TiO2 under Solar-Light Irradiation with XPS, Solid-State NMR, and DFT Calculations, J. Am. Chem. Soc. (2013) 135, 4, 1607–1616

II. Why do we need this module?

♦ From the atomic level, explain the experimental results

Case Three: Graphene/Cu composite with enhanded Thermal performance

Graphene can enhance the thermal conductance for Cu system

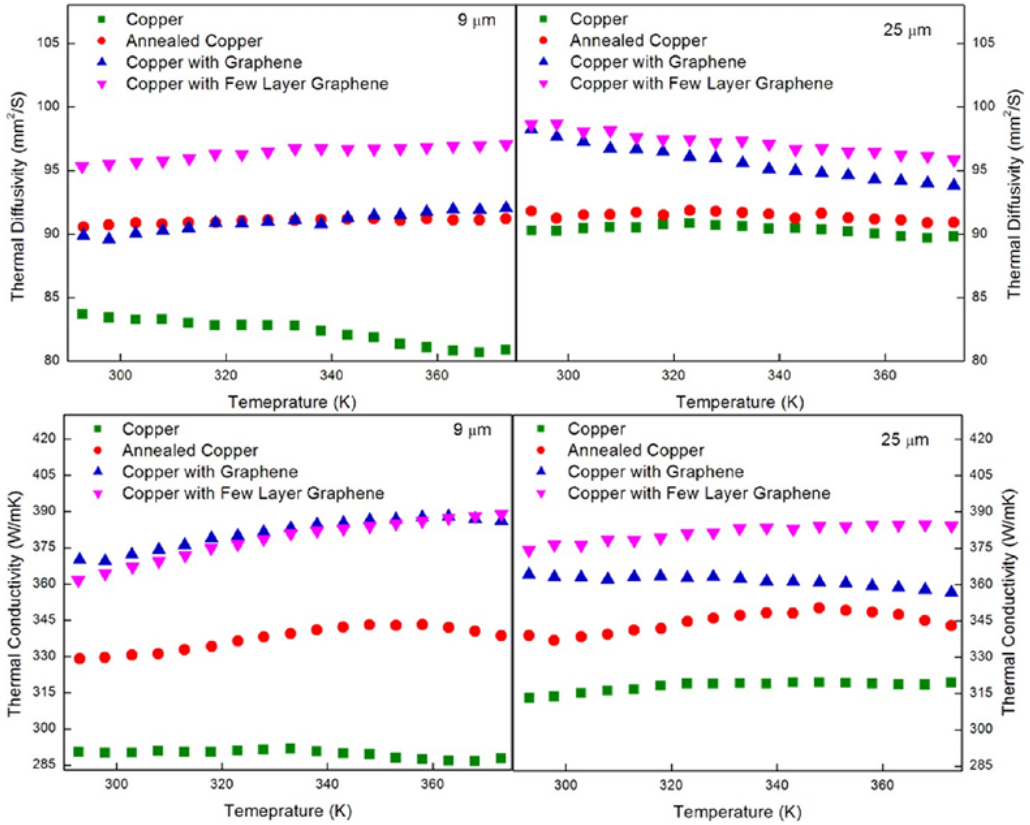
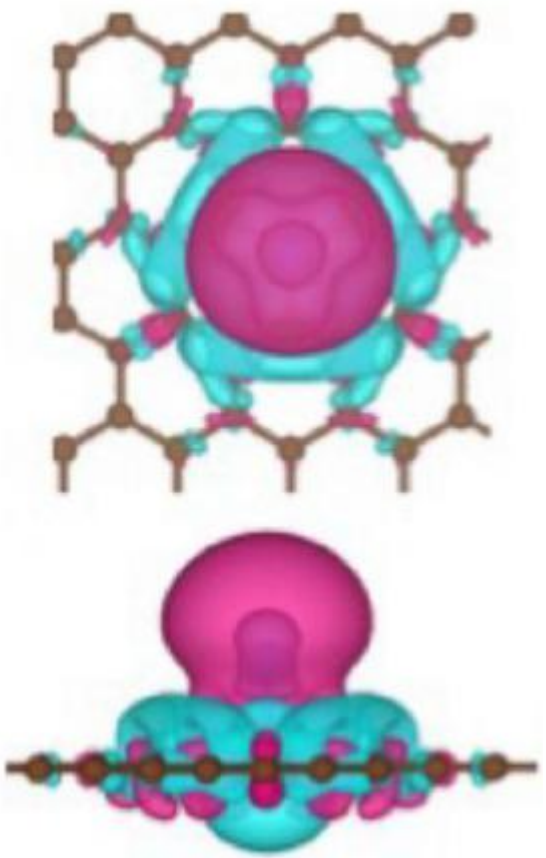


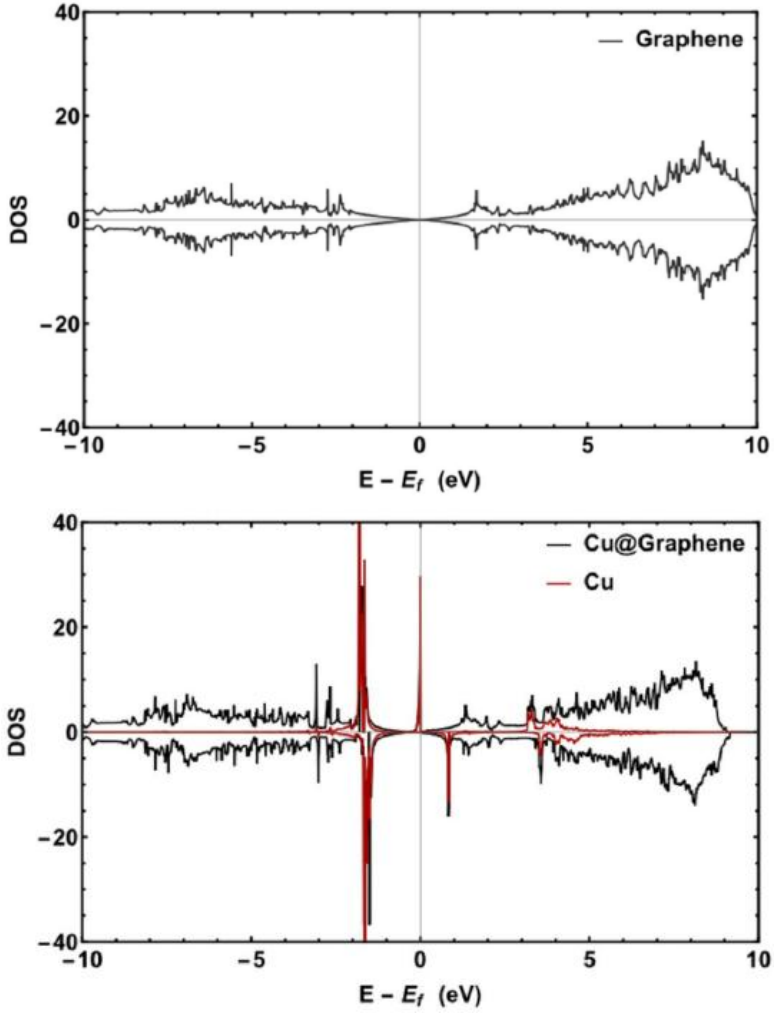
Figure 2. Thermal diffusivity and thermal conductivity of graphene-coated copper films. Thermal diffusivity of reference Cu film, annealed Cu, Cu with CVD graphene, and Cu with CVD FLG (top panels). Thermal conductivity of reference Cu film, annealed Cu, Cu with CVD graphene, and Cu with CVD FLG (bottom panels). The data are shown for Cu films with $H = 9\ \mu m$ and $H = 25\ \mu m$. Note that CVD of graphene and FLG results in stronger increase in the apparent thermal conductivity of graphene−Cu−graphene samples than annealing of Cu under the same conditions.

Pradyumna Goli, Hao Ning, Xuesong Li, *et al.*, Thermal Properties of Graphene–Copper–Graphene Heterogeneous Films, Nano Lett. (2014) 14, 3, 1497–1503

Case Three: Graphene/Cu composite with enhanded Thermal performance

More hybrid states between Graphene and Cu result in the enhanced thermal conductance



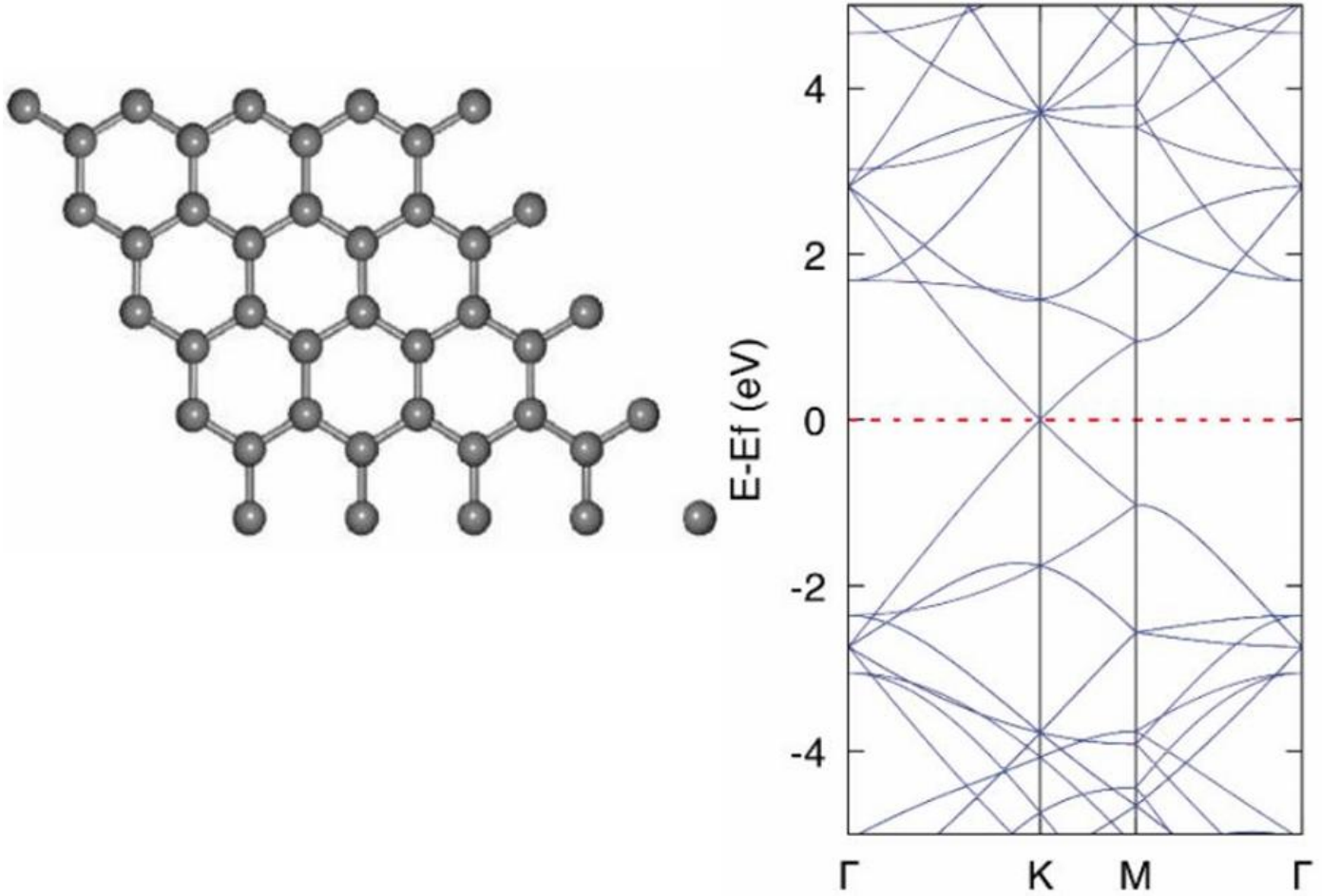


Aligholami, M., Akbari, M., Morad, R. et al., Enhanced thermal conductivity of Cu Grafted graphene-$C_2H_6O_2$ based nanofluids by laser ablation for potential application as coolants in data centers. Sci Rep (2025) 15, 17679

II. Why do we need this module?

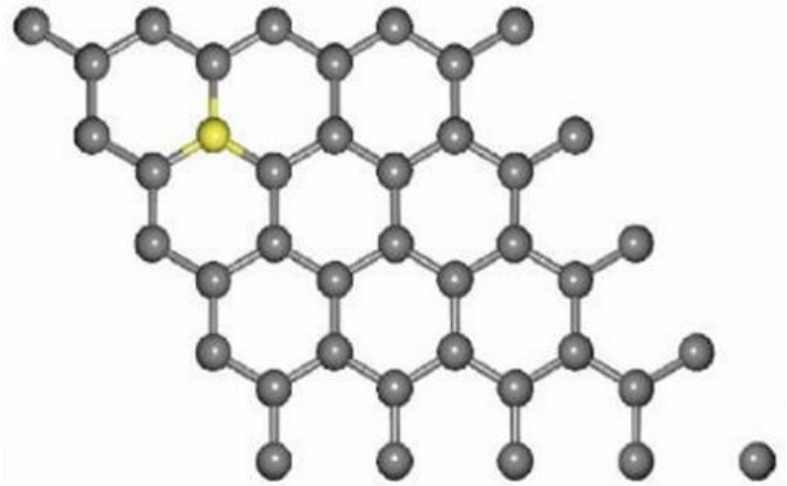
♦ From the atomic level, modify the electronic properties of materials

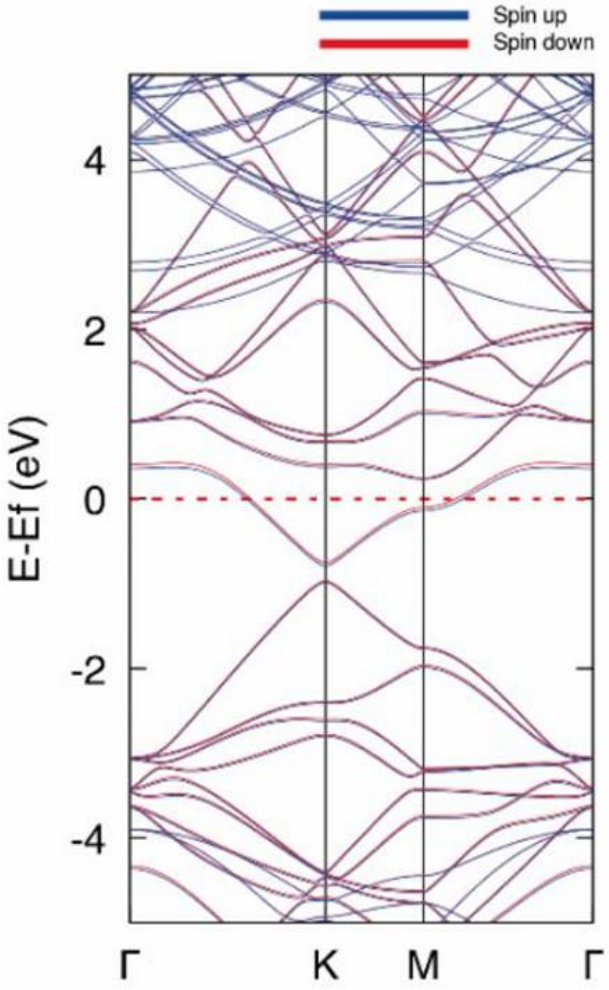
Case One: Engineering the band structure of Graphene



A. J. W. T. Nenohai, F. M. Santana, R. Asih, et al., Effect of N doping on Electronic Band Structure and Magnetic Moments in Graphene Sheets: Density Functional Theory Study, Journal of Physics: Conference Series 2780 (2024) 012009

Case One: Engineering the band structure of Graphene

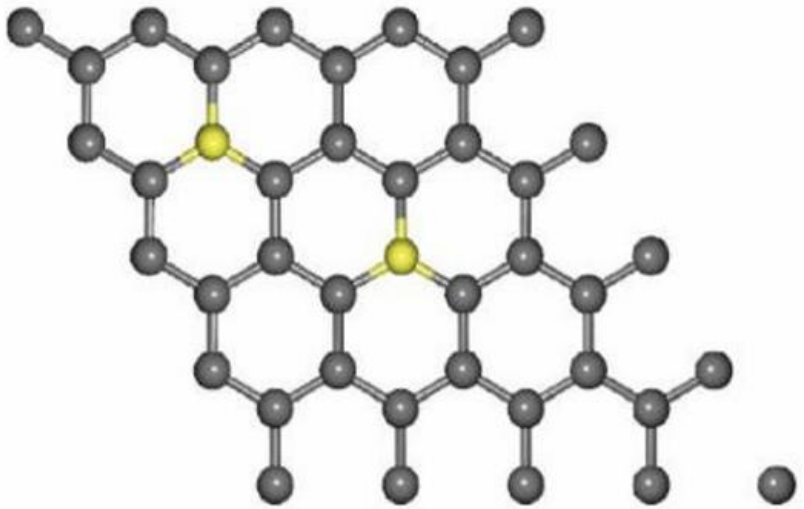


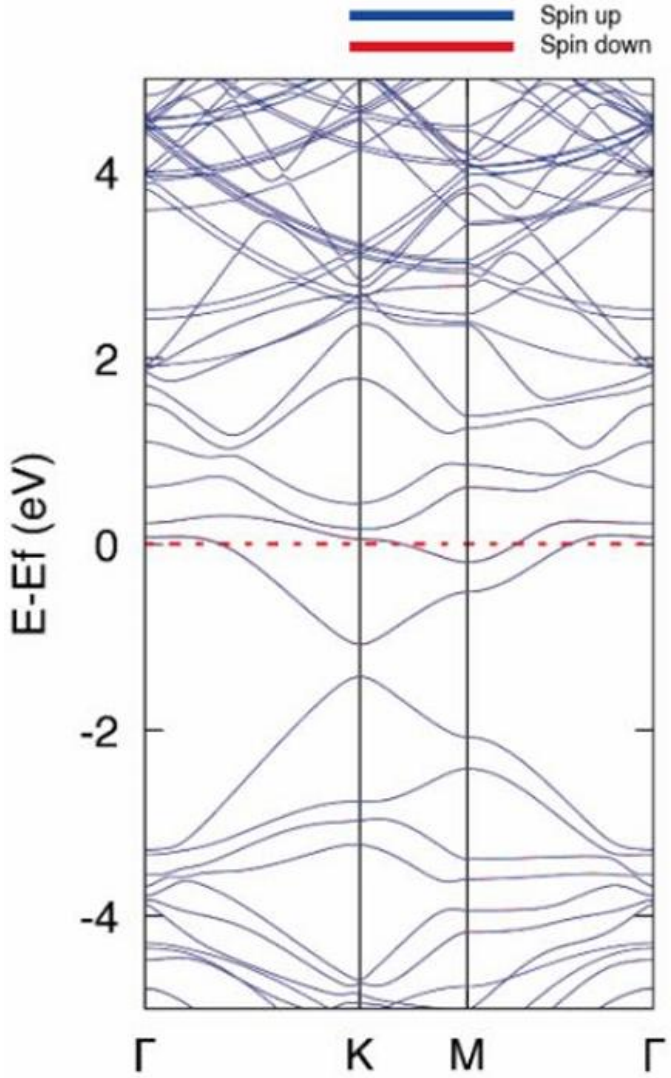


Gap opening + spin degenerancy released with N doping

A. J. W. T. Nenohai, F. M. Santana, R. Asih, et al., Effect of N doping on Electronic Band Structure and Magnetic Moments in Graphene Sheets: Density Functional Theory Study, Journal of Physics: Conference Series 2780 (2024) 012009

Case One: Engineering the band structure of Graphene





Gap increases with N doping percentage

A. J. W. T. Nenohai, F. M. Santana, R. Asih, et al., Effect of N doping on Electronic Band Structure and Magnetic Moments in Graphene Sheets: Density Functional Theory Study, Journal of Physics: Conference Series 2780 (2024) 012009

II. Why do we need this module?

♦ From the atomic level, modify the electronic properties of materials

Case Two: Tuning charge conductance of MoS_2 by strain

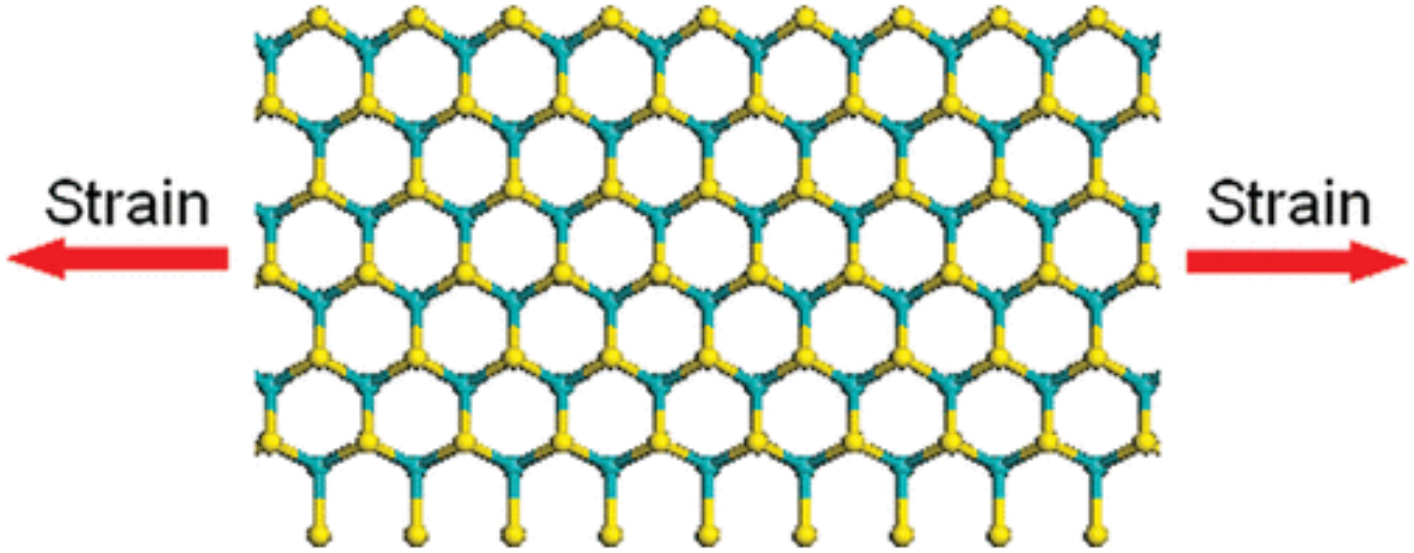
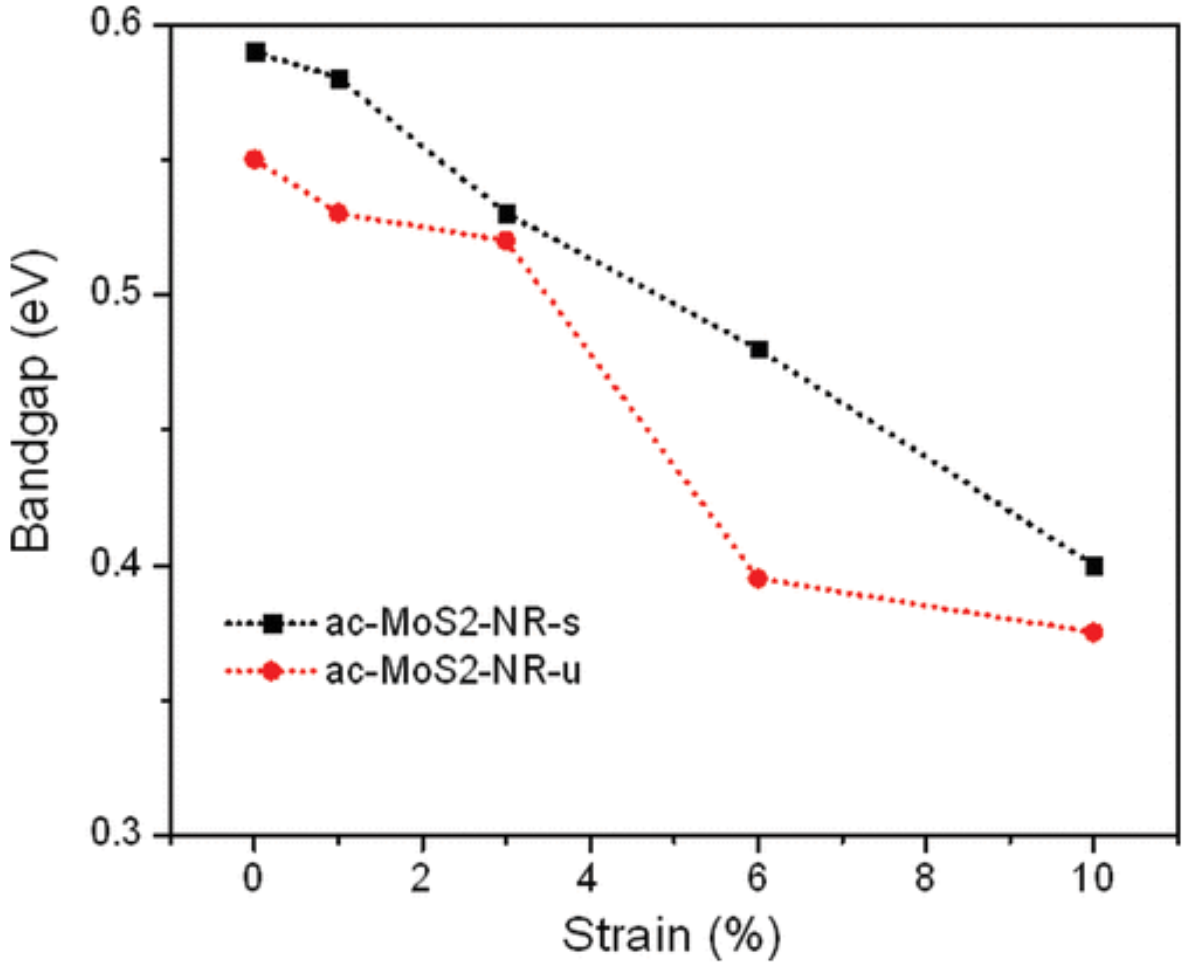


Figure 2. Representative scheme of a MoS_2 nanoribbon with the applied strain along its axis.

Hui Pan and Yong-Wei Zhang, Tuning the Electronic and Magnetic Properties of MoS_2 Nanoribbons by Strain Engineering, J. Phys. Chem. C 2012, 116, 21, 11752-11757

Case Two: Tuning charge conductance of MoS_2 by strain

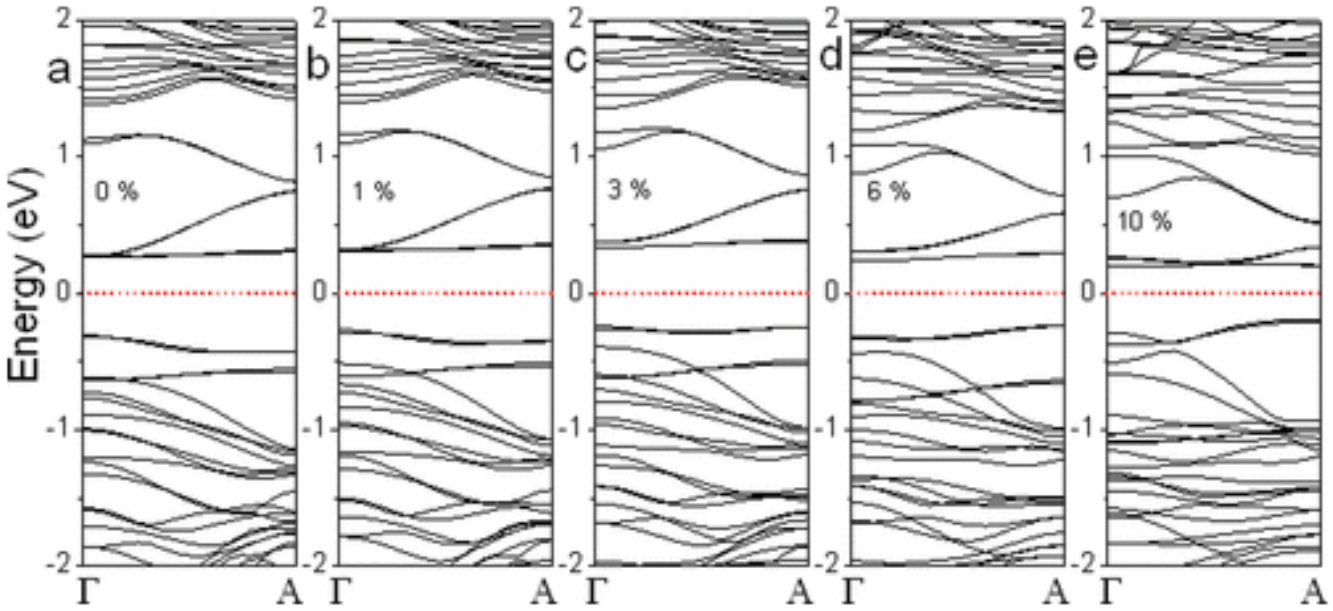


Applied strain brings gap narrowed down

Figure 5. Calculated bandgaps of armchair MoS_2 nanoribbons as a function of applied strain.

Hui Pan and Yong-Wei Zhang, Tuning the Electronic and Magnetic Properties of MoS_2 Nanoribbons by Strain Engineering, J. Phys. Chem. C 2012, 116, 21, 11752-11757

Case Two: Tuning charge conductance of MoS_2 by strain



More states shows up within the gap as strain applied

⬇

These states assist charge moving with lower energy

⬇

Charge conductance enhanced with low voltage

Figure 4. Calculated band structures of armchair MoS_2 nanoribbon with symmetric edge structure (ac-MoS_2−NR-s) under various strain (a) 0, (b) 1%, (c) 3%, (d) 6%, and (e) 10%.

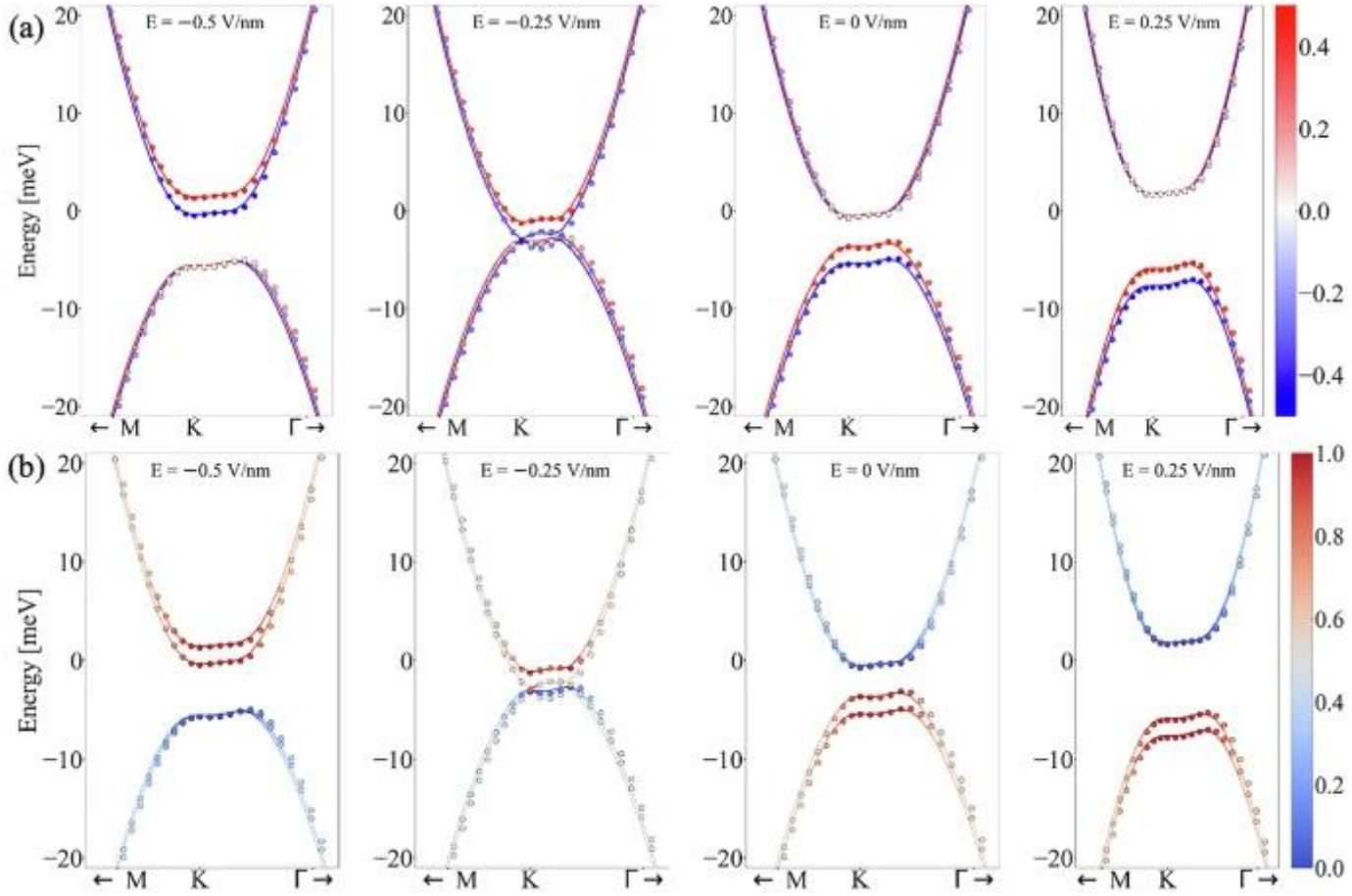
Hui Pan and Yong-Wei Zhang, Tuning the Electronic and Magnetic Properties of MoS_2 Nanoribbons by Strain Engineering, J. Phys. Chem. C 2012, 116, 21, 11752-11757

II. Why do we need this module?

♦ From the atomic level, modify the electronic properties of materials

Case Three: Bilayer Grpahene/WSe_2 for Quantum Entanglement tuning

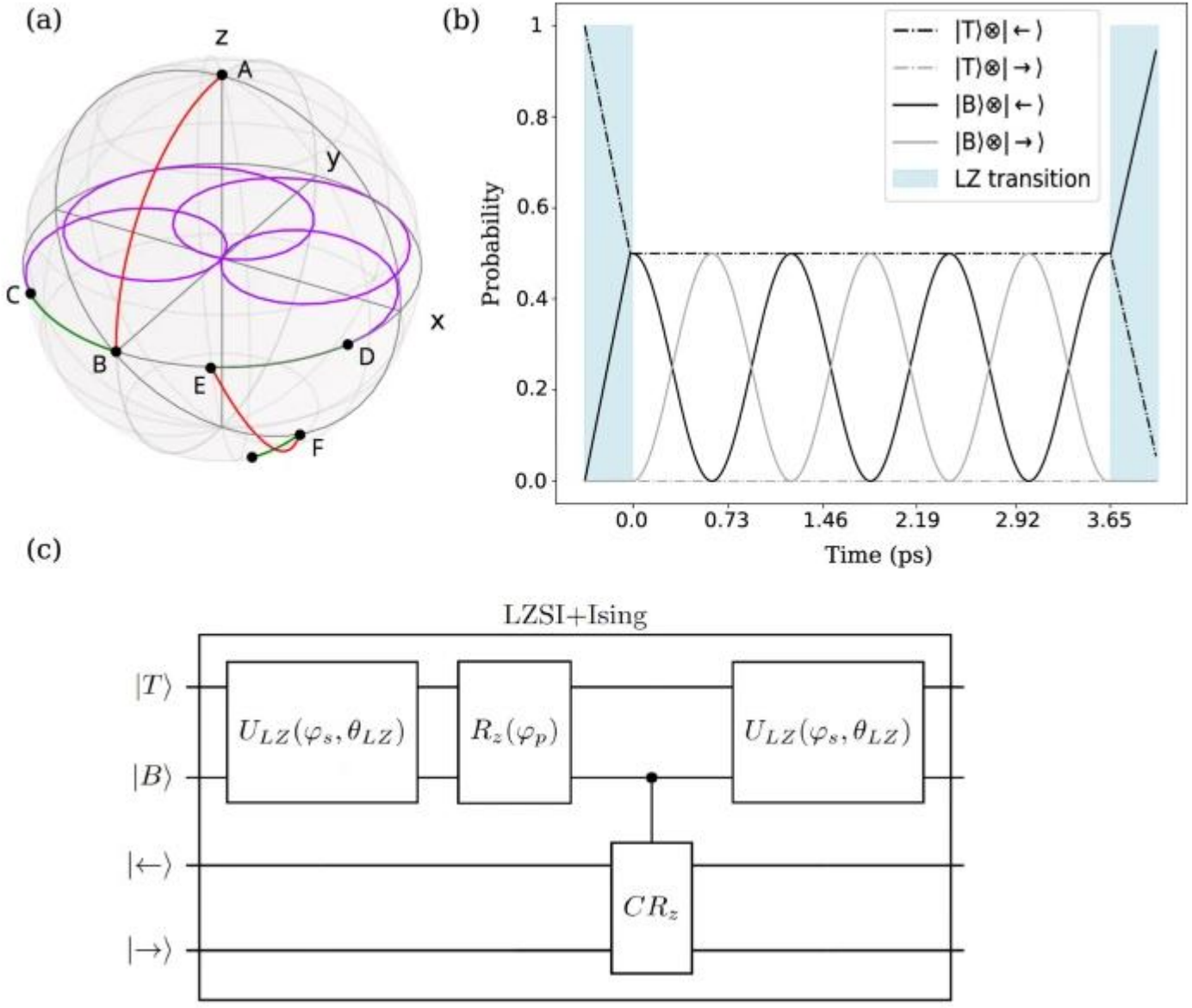
Spin degeneracy released on different Graphene layer with different E field in BLG/WSe_2 system



Ge H., Koopmann P., Mrcarica F., *et al*., Ab initio simulation of spin-charge qubits based on bilayer graphene-WSe_2 quantum dots, npj 2D Mater Appl, (2025) 9, 47

Case Three: Bilayer Grpahene/WSe_2 for Quantum Entanglement tuning

Fast tuning the spin switching achieved in Quantum computing circuit

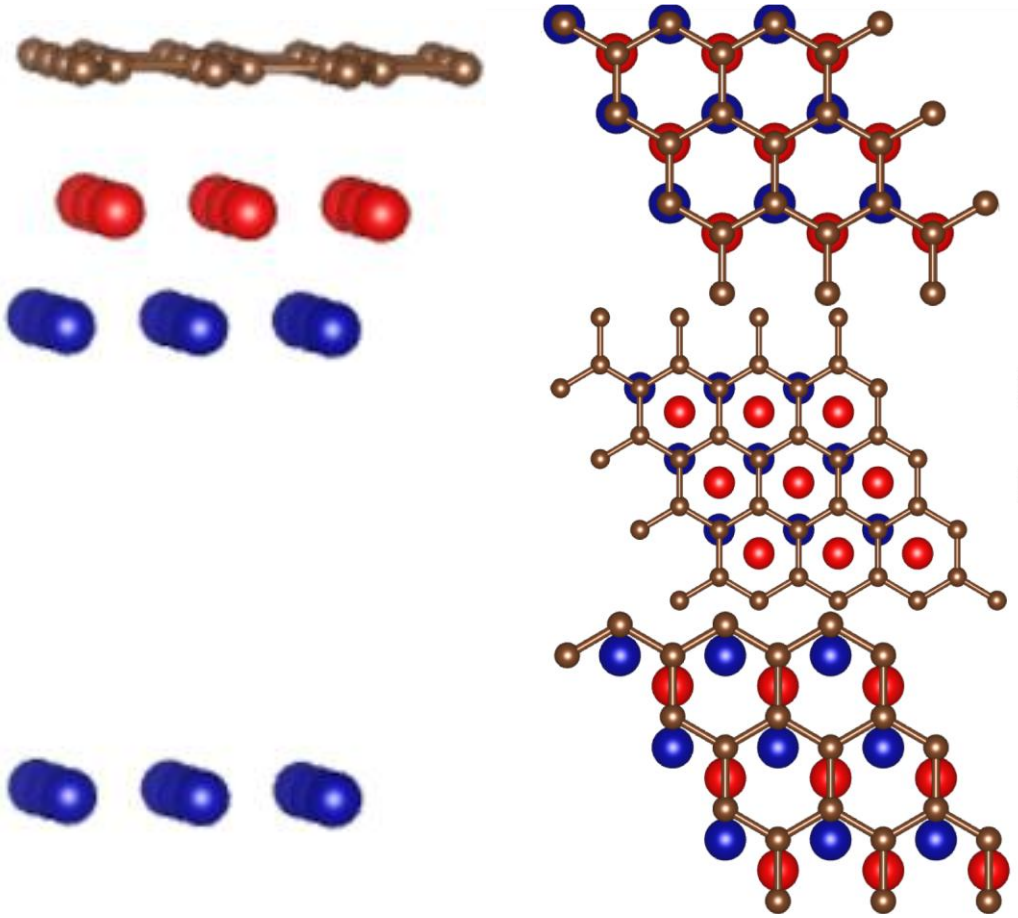


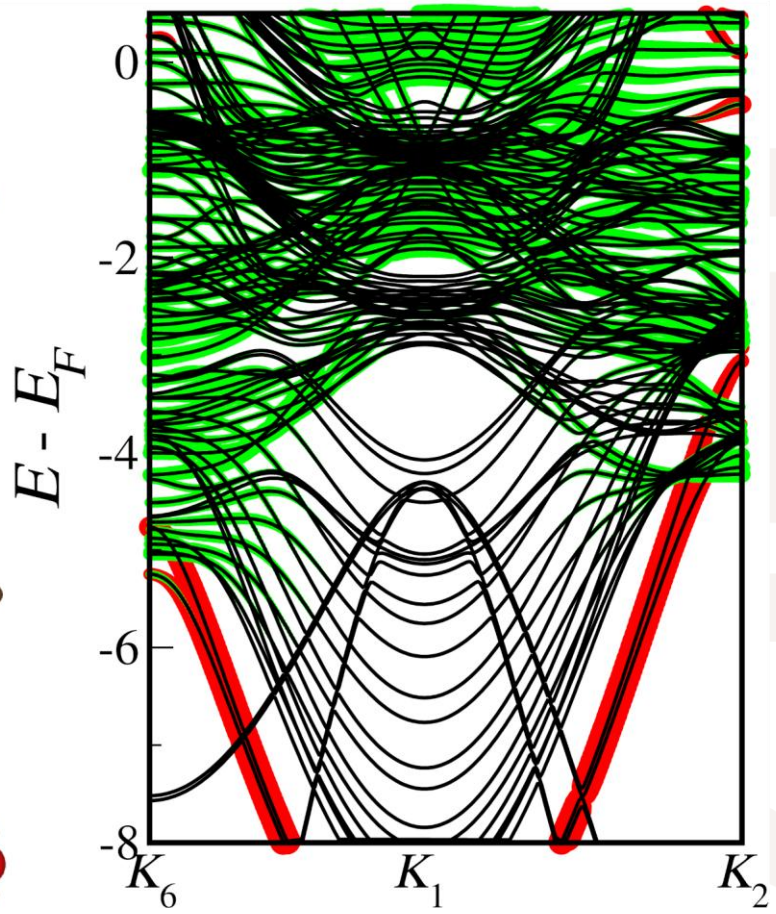
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II. Why do we need this module?

♦ From the atomic level, predict the properties of materials

Case One: Orbital Torque in Graphene/Co for memory chips



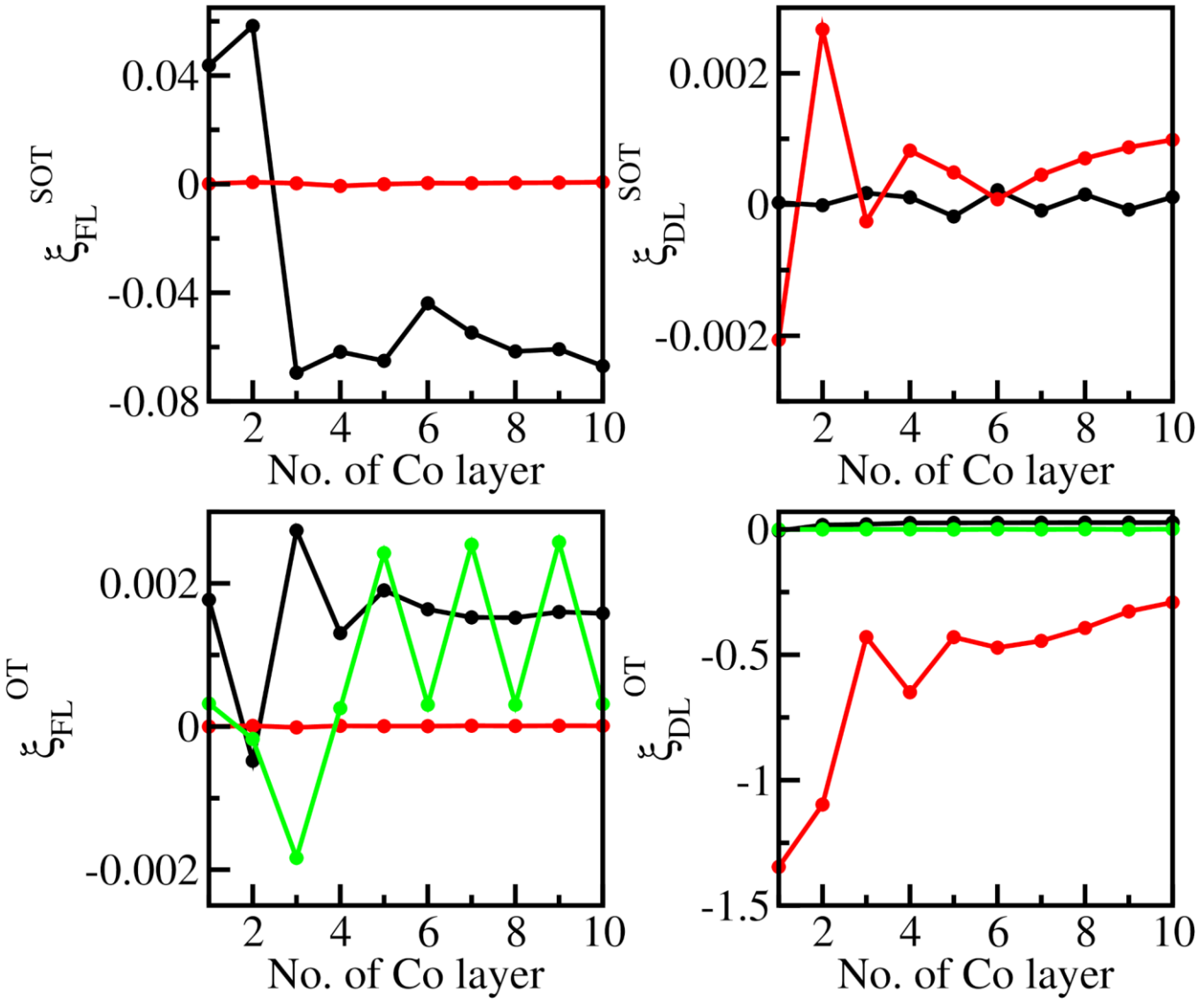


Band structure of Graphene/Co system

Case One: Orbital Torque in Graphene/Co for memory chips

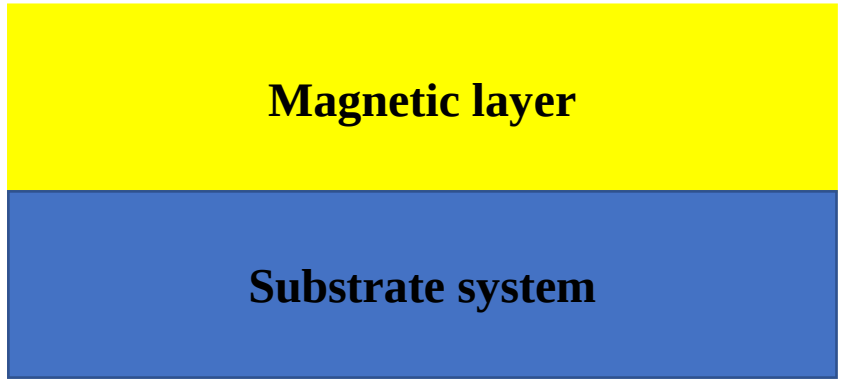
Ratio of Torque/Conductivity reaches almost 50% with thickness

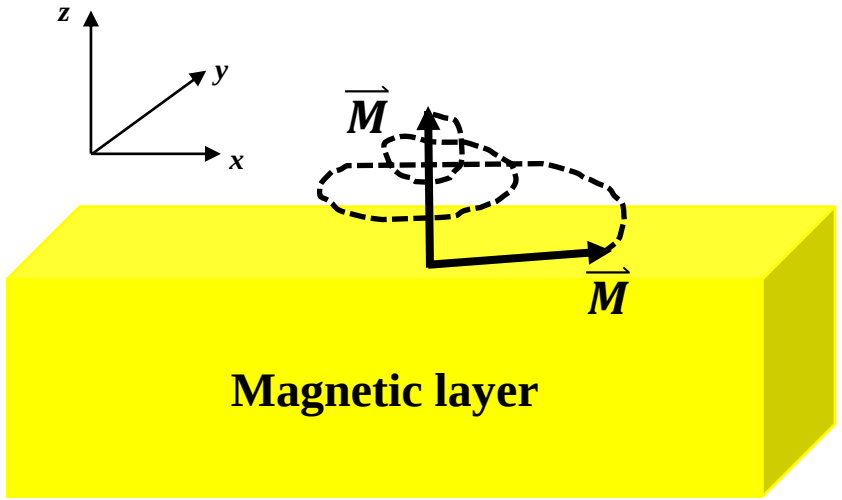
Magnetisation switching with pretty low power



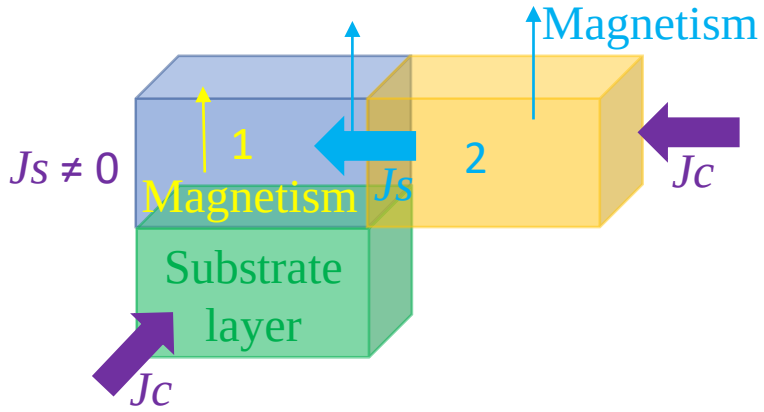
OT ratio much more larger than SOT one in Graphene/Co system

Case One: Orbital Torque in Graphene/Co for memory chips

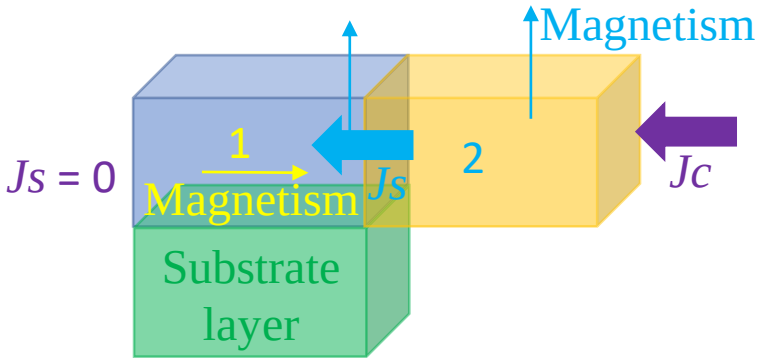




Spin Orbit Torque



Initial State



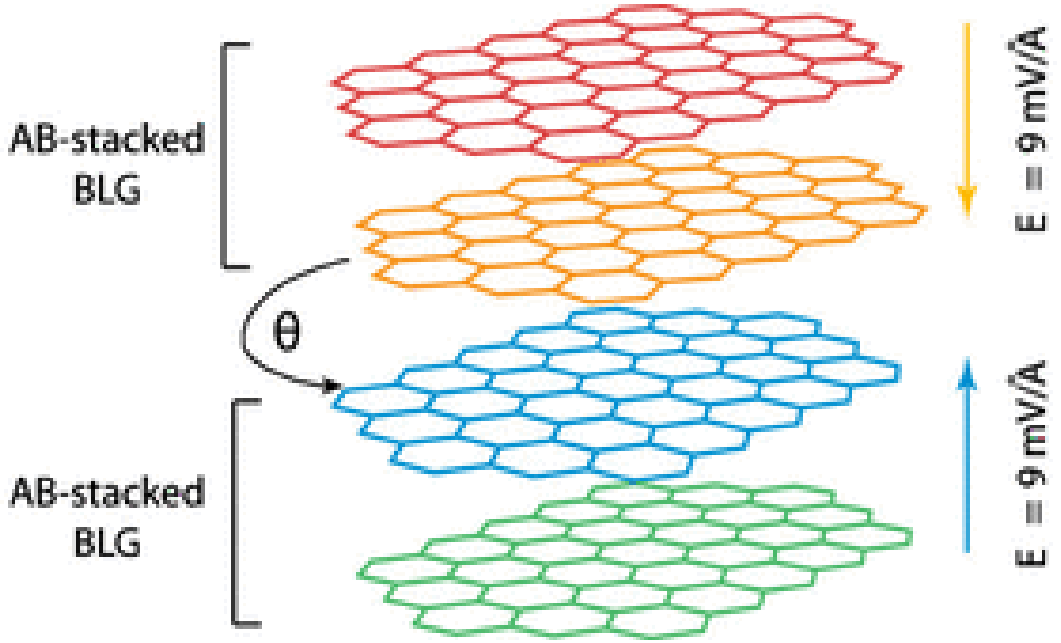
Final State

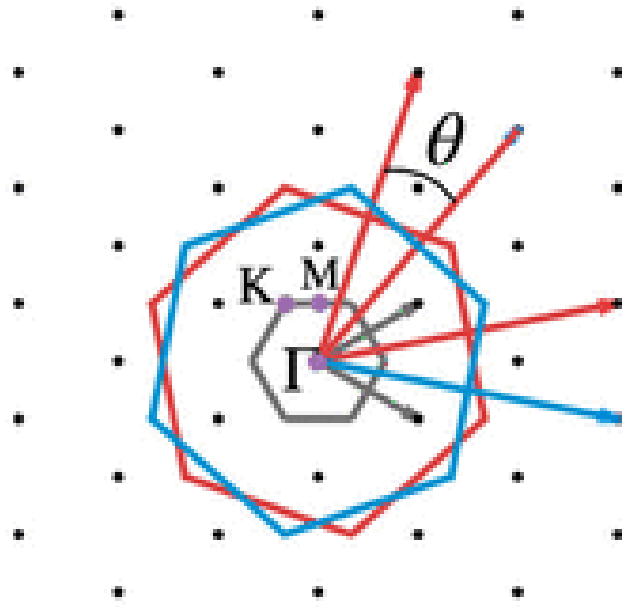
Fast Signal switching (0.4 nanoseconds) in spintronic devices with the application for memory devices

II. Why do we need this module?

♦ From the atomic level, predict the properties of materials

Case Two: Twisted bilayer Graphene for superconductivity





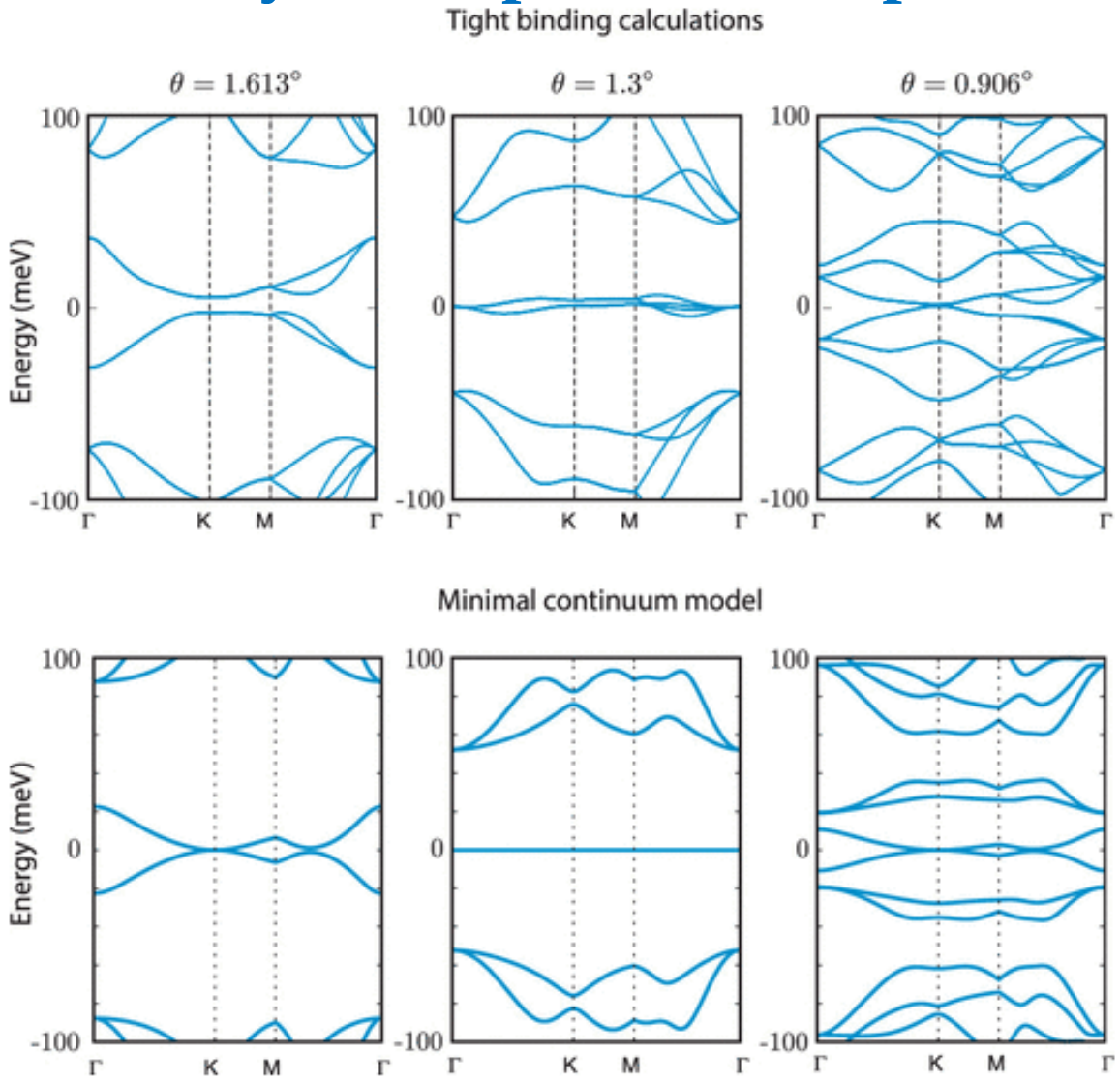
Twisted bilayer graphene with different stacking

Fatemeh Haddadi, QuanSheng Wu, Alex J. Kruchkov and Oleg V. Yazyev, Moiré Flat Bands in Twisted Double Bilayer Graphene, Nano Lett. 2020, 20, 4, 2410-2415

Case Two: Twisted bilayer Graphene for superconductivity

Flat band achieved with different twisted angle

Approaches superconductivity



Xi'an Jiaotong-Liverpool University
西交利物浦大学

Fatemeh Haddadi, QuanSheng Wu, Alex J. Kruchkov and Oleg V. Yazyev, Moiré Flat Bands in Twisted Double Bilayer Graphene, Nano Lett. 2020, 20, 4, 2410-2415

II. Why do we need this module?

- From the atomic level, explain the experimental results
- From the atomic level, modify the electronic properties of materials
- From the atomic level, predict the properties of materials

Guide experimentalist to focus on specific materials with special properties

- Have a better understanding of the origin of materials with different properties
- Fabricate device with the materials for difference purpose

III. How is this module delivered?

52 Lectures
4 hours/week × 13weeks

Theory and some case study

Some fundamental programming skill

+

24 Lab
4 hours/week × 6weeks

Reproduce the theoretical results delivered on the lecture

Practice some simple calculations with results analysis

III. How is this module delivered?

Assessment for this module

Reproduce the theoretical results for the graphene system delivered on the lecture

Submit a report and provide related analysis, based on the lab work

6-week lab work

IV. Overview of the content framework of this module

Lectures

- ♦ Week 1: Brief introduction to Quantum Mechanics
- ♦ Week 2: Crystal structure in both real and reciprocal spaces
- ♦ Week 3: Tight Binding Method and band structure calculation
- ♦ Week 4: Tight Binding Method and Density of State calculation
- ♦ Week 5: Tight Binding Method and Spin Texture
- ♦ Week 6: Tight Binding Method and Charge Conductivity
- ♦ Week 7: Deduction of Charge Conductivity Kubo-Greenwood and NEGF methods
- ♦ Week 8: Programming skill based on Fortran: Parallelization Technique
- ♦ Week 9: Density Functional Theory
- ♦ Week 10: Introduction to DFT Code: OpenMX
- ♦ Week 11: Band structure, DOS calculation with OpenMX
- ♦ Week 12: Hamiltonian and Overlap matrices from OpenMX
- ♦ Week 13: Example Study based on OpenMX+Numerical coding: SOT in Graphene/Co System

IV. Overview of the content framework of this module

Lectures

♦ **Weeks 8 - 13 with 4 hours/week: Coding work to reproduce the band structure, DOS, charge conductivity of pristine Graphene with Python/Matlab or any other programming**



That's all for this session

Thank you for your attention and great participation